(11) You are given a binary search tree with N elements and asked to return the Kth largest and Kth smallest element in the BST. What is the time complexity of your solution?

(12) You are given an input string and asked to write a code snippet to return the run length encoded representation of that string. For example, if you are given the string aaaccefhhh, your function should return the string a3c2e1f1h3.

(13) You are given a binary tree that contains N integers and asked to verify whether it is a binary search tree. What is the time complexity of your solution?

(14) You are given a singly linked list that you don't know the length of. You are asked to write a function that returns the middle element of the list. You can assume that there are an odd number of nodes on the linked list so that you have a unique middle element.

(15) We are all familiar with the dynamic memory allocation routines—*malloc* and *free*. The *free* function typically takes a single parameter, which is the pointer to the memory to be freed. How does the *free* function know what is the size of memory to be freed?

(16) You are given a continuous stream of integers. At any time during the reception of the stream of numbers, you can be asked to give the median of the numbers seen so far. Write an algorithm for this online median problem. What is the time complexity of your solution?

(17) Given an array A of N integers, write a function that prints all sets of numbers A, B, C and D belonging to array A such that A+B = C+D.

(18) Given an adjacency list representation of a graph G with V vertices and E edges, write an algorithm to check if it is a bi-partite graph. In a bi-partite graph, its vertex set can be partitioned into V and V' such that all edges go between V and V' only.

(19) Can you explain what happens when your C program receives a 'segmentation fault' error?

(20) Can you explain the difference between procedural and functional languages? Is Java a functional language?

(21) How are virtual functions implemented in C++ language? If you want to implement virtual functions in C, how would you go about it?

(22) If you create an empty struct in C, and invoke the sizeof() operator on it, what would be the size returned? If you do the same in C++, what would be the size returned? Would they be the same? If not, why not?

(23) When would you use reader-writer locks vs mutex locks? Do reader-writer locks have greater performance overheads compared to mutex locks?

(24) What is the processor architectural level primitive needed if you want to implement a mutually exclusive lock in the operating system?

(25) Your application consists of multiple application threads that are accessing/updating a global shared counter. Hence, some means of serialising the updates is needed to ensure atomicity of the update. The global counter is an integer variable. How would you ensure that the variable is updated in an atomic and consistent manner? Do you need to use a mutex lock to protect the variable? If not, how would you carry out the access/update?

(26) You have a hash table abstract data type defined, which supports the insertion, deletion and looking-up of an item. The hash table interfaces are thread-safe such that insert/delete/search are protected by means of a mutex lock associated with that hash table. For your application, you need to instantiate two hash tables, namely, the source hash table and destination hash table. Based on certain conditions, you will need to move items from the source hash table to the destination hash table atomically. You are asked to implement a *transfer_item* function using existing interfaces of insert/delete/search. How would you implement this function? If not possible, explain why.

(27) You are asked to implement an abstract data type dictionary which supports the look-up/insert/delete operations. What would be the choice of your data structure for implementing the dictionary? When would you prefer a hash table rather than a red-black tree or a binary search tree?

(28) Do all modern processors support out-of-order execution? Can you name a processor that is an in-order processor?

(29) What is meant by precise interrupts?

(30) When the Linux kernel is running, is it possible to take a live crash dump of the system? If you have to design a mechanism to perform a live kernel dump, how would you go about it?

If you have any favourite programming questions/ software topics that you would like to discuss on this forum, please send them to me, along with your solutions and feedback, at s*andyasm_AT_yahoo_DOT_com*. Till we meet again next month, Happy New Year and happy programming! END

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